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PI Name:	Schreckenghost, Debra M.E.E.		
Project Title:	HCAAM VNSCOR: Enhancing Situation Awareness of Automated Procedures Using Adaptive Multimodal Augmented Reality Displays		
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Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
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Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
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Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Task Description:

This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR).
Future deep space missions will present new challenges for crew, and increased risks to human performance due to the stress, fatigue, radiation exposure, and isolation that characterizes these missions. In addition, crew will no longer be able to depend on timely support from Mission Control due to distance from the Earth, but will have to work autonomously, while maintaining high performance. Mission Controllers may not be available to answer questions, check system status, assist with procedures, monitor for errors, or troubleshoot problems. Greater crew autonomy will increase dependence on automated systems, and design of these automated systems must be driven by sound human-system integration standards and guidelines in order to ensure mission success. Historically, crew have had very limited dependence on automated systems, thus crew will be faced with a new way of working that may put situation awareness (SA) at risk. We must develop methods for promoting good situation awareness in the automated systems that will most certainly be part of future deep space vehicles and habitats.

Procedure automation is a promising technology for reducing crew workload. We define procedure automation as technology that automates the selection or execution of procedural tasks. Structuring the work of automation according to human procedures should improve the transparency of automation actions. This approach provides a means for establishing common ground about ongoing tasks to improve operator understanding of automation behavior.

New technologies such as adaptive, multimodal, augmented reality displays can offer the benefits of information presentation tailored to meet the needs of each crewmember, taking into consideration the current state of that crewmember (e.g., sleep-deprived, high workload), as well as the current state of his/her environment and ongoing activities (e.g., emergency situation, time-critical operations).

We combine technology for procedure automation with technology for augmented reality multi-modal (ARMM) user interfaces using Microsoft Hololens head-mounted display to provide a virtual task assistant to assist crew in performing procedural work. This virtual task assistant identifies procedures to perform, aids performing actions in crew procedures, and summarizes actions taken by the human-automation team to assist crew in preparing for tasks and taking over tasks from other team members.

Four studies were conducted to evaluate the effects of a virtual task assistant combining procedure automation with augmented reality multi-modal (AARM) user interfaces on human task performance. These studies will achieve the following aims:

Aim. Determine best methods to improve situation awareness and improve crew autonomy when using a virtual task assistant to prepare for and perform manual maintenance.

Aim. Determine best practices to improve crew performance when using virtual task assistant to aid in task transitions (task handovers, automation interruption).

Aim. Determine best methods for display of procedural work to improve crew autonomy and satisfaction

The proposed work addresses a number of gaps in the Human Research Program Human Factors and Behavioral Performance risks. This project will provide guidelines for designing effective human-automation systems and evaluate human-automation performance for exemplar procedure automation systems. This project also will provide guidance for the application of multi-modal and adaptive displays and control to Human-Computer Interaction (HCI) design for long duration operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Technologies for virtual task assistance are increasingly available in everyday life. One of the most common is voice enabled assistance, like Siri and Alexa, that aid some activities of daily living. And augmented and virtual reality technologies are becoming mainstream, with the introduction of new devices such as Microsoft HoloLens 2 and Apple's Vision Pro.
The Virtual Intelligent Task Assistant (VITA) project is leveraging augmented reality platforms and toolkits for augmented reality apps to develop a virtual task assistant that can be used to assist users with procedural task work on the job. Our technical approach is innovative in that new procedural tasks can be supported without custom software development. Our experimental research is distinguished by investigating effective task assistance for maintenance or assembly tasks where hands-free operation of task assistance is beneficial. For the first study we investigated best techniques for using augmented reality task assistance when assembling small rovers that are held in the hands during assembly. Subsequent studies investigate other aspects of using augmented reality for assembly tasks, including use of 3D images in procedures and virtual notification for task milestones and transitions.

This technology and associated research findings have potential benefit to NASA for the assembly, maintenance, and repair of aircraft, spacecraft, habitats, and robotics. This technology and associated research findings also have broader potential benefit for any organization performing assembly and maintenance procedural work. This includes assembly and maintenance of drilling equipment for the oil and gas industry, equipment used in chemical processing plants, and maintenance and repair of commercial aircraft.

I. VITA Study in HERA Campaign 7
The aim of the Virtual Intelligent Task Assistant (VITA) study in NASA Human Exploration Research Analog (HERA) Campaign 7 (C7) is to determine best methods for the display of procedural work to improve crew autonomy and satisfaction. Specifically, this study compares crew performance and satisfaction on rover assembly tasks similar to those used in HERA C6 but with procedures that vary the level of detail in text instruction and the visual aids, to improve support for flexible execution of actions. These differences are illustrated in existing VITA procedures for rover assembly and disassembly.

A new version of the VITA task assistant was developed for use in HERA C7. The new task assistant is an augmented reality application developed in Unity. This application runs on the augmented reality headset Microsoft's HoloLens 2, instead of the HoloLens 1 used in HERA C6. The VITA augmented reality display was redesigned, including changing from gaze interaction used in HERA C6 to voice interaction for HERA C7.

Task Progress:

To address our aim of best methods for the display of procedure work, we investigate procedures that vary the visual aid and the level of detail in text. Specifically, we compare performance with figures displaying Photographs to 3D models with diagrams in figures. We also look at the level of detail in text instructions, comparing procedures with and without more detailed text instructions.

The VITA study in HERA C7 had a total of 16 participants. Participants were selected by HERA to be astronaut-like. Participants were trained to use VITA prior to each HERA mission. Data collection includes situation awareness, workload, usability, and task timing at the end of each session. A survey was administered at the end of each session (called the end of condition survey) and after the final VITA session (called the end of mission survey). These surveys include questions about user satisfaction and perception of autonomy when using different procedure styles will be added to the end of mission survey.

During the reporting period, the VITA project met all HERA C7 delivery milestones. This included training the crews of C7M2-4, and supporting VITA sessions as needed during C7M2-4. It also includes analysis of data collected during HERA C7, which is ongoing at the time of this report.

Preliminary findings are summarized on human performance using the different visual aids and level of detail during the VITA study in HERA C7. The Bedford workload scale was administered at the end of each assembly session. For the first time an assembly task is performed, the Photograph condition has a median workload of 6, while the 3D condition has a much lower median workload of 3.5. When the procedure is repeated; however, the median workload is the same at 3.0 for both visual aids. Some workload differences may be due to task differences.

System usability is measured using the System Usability Scale (SUS). SUS was assessed at the end of each assembly session. Mean usability ratings for the Photograph and the 3D model conditions are comparable for both the first and second repetition of the procedure. Means values are above 70 for all conditions.

Situation awareness (SA) was assessed by the participant at two points in the assembly task, one halfway through the task (SA1) and one just after the task was finished (SA2). The participants rated SA from a low of 1 to a high of 7. For the interim rating SA1, the median SA for the Photograph condition is lower at 5 while the median SA for the 3D model is 6. The final rating SA2; however, is similar for both visual aids, with a median value of 5-5.5 for the first repetition, and a median value of 6 for the second repetition. All SA ratings are moderately high.

Participants were asked which visual aid they preferred. Ten of 16 crew participants preferred using the 3D model. A key reason given was that the 3D model provides more information more quickly. Some participants also felt the 3D model was easier to use. Additionally, the 3D model shows the relationship between the parts to be assembled and makes this relationship available at all steps of the assembly. Four of 16 participants preferred figures using Photographs. These participants felt photographs were straightforward to use and captured more detail than the 3D model. Finally, two of 16 participants had no preference but mentioned that they liked having both options for different situations.

Participants also were asked which level of text detail they preferred: Expanded text or Collapsed text. Ten of 16 participants preferred the Expanded text because it gave them additional information complementary to the visual aids or not available in the visual aids. Some also felt the text instruction provided useful context in interpreting the visual aids. Two participants preferred having less text detail, feeling the visual aids were sufficient to perform the task. Three participants felt the text instruction was useful sometimes, such as the first time they built a rover configuration, or if an instruction was particularly complex.

II. Laboratory Study of VITA Support for Task Transitions

A laboratory study was performed investigating aid from the VITA task assistant for task transitions. This study aims to determine the best practices to improve crew performance when using the VITA virtual task assistant to aid in task transitions, such as handling the interruption of a manual task by an automated task. In this study, the user is responsible for the manual assembly of the rover gripper. Concurrently they are responsible for supervision of an automated task (e.g., the automated startup of a habitat system). Supervisory tasks include monitoring the progress on the startup, and intervening in the automated task when automation pauses.

The participant monitors progress on automated tasks using virtual notices displayed in the VITA task assistant. These notices are issued when task milestones are achieved, corresponding to changes in system states or environmental levels.

This study evaluated two conditions – with and without virtual notices from the automated task displayed in the task assistant. The condition With Virtual Notices displayed notices in the HoloLens. The condition Without Virtual Notices did not display notices in the HoloLens or in a Tablet. Instead, the participant monitored automated task progress and the need for intervention by periodically checking the procedure display on the laptop. User intervention with the automation was done for both conditions using PRIDE procedures displayed in a browser on a laptop. Task performance was compared with and without virtual notices in the VITA task assistant. This is a within-subjects study.

The VITA laboratory study had 16 participants. The study was conducted in the Human Factors Engineering Laboratory (HFEL) onsite at NASA Johnson Space Center (JSC) using subjects from the Human Test Subject Facility (HTSF). Participants were trained prior to the study session. After training, each participant assembled the gripper, but without responsibility to supervise automation. Then a second gripper assembly was performed, during which the participant was responsible for supervising automation. During the second gripper assembly, participants performed supervisory tasks both with and without virtual notices from the automated task. Sessions were counterbalanced for these notification conditions.

During the reporting period, the VITA project completed data collection for the lab study and began analysis of these data, which is ongoing at the time of this report.

Preliminary findings are summarized by comparing human performance on the first gripper assembly using the different visual aids during the VITA study in HERA C7 and the VITA lab study. This comparison was performed to gain insight into differences in the visual aids used for each study. The HERA C7 visual aids combine a 3D model of the gripper with annotated 2D images of the model. The VITA lab study replaces the 2D images used in HERA by overlaying the annotations on the 3D model (“enhanced 3D model”).

Situation Awareness was measured during the assembly task. Two SA ratings, from a low of 1 to a high of 7, were assessed by the participant, one halfway through the task (SA1) and one just after the task was finished (SA2). Both

	studies have a mean value at SA1 just below 6 and a median value of 6 for the HERA study and 6.5 for the lab study. The ratings at SA2; however, show larger differences. Both the mean and median values using the modified visual aids for the Lab Study are essentially the same at both SA1 and SA2. However, the SA2 ratings for the HERA C7 study have both a mean and median value of 5. The mean value is one point lower than SA1 and the median value is 1.5 points lower than SA1. We are investigating the possible causes of this loss. Situation awareness was also assessed using the Modified Situation Awareness Rating Technique (SART) to measure SA. The SART technique assesses situation awareness in three areas: task demand, crew resources, and crew understanding. Task Demand: Both the mean and median Task Demand ratings are over a point lower for the VITA lab study, indicating a less demanding task. These differences in the Task Demand scale may be affected by the design changes in the visual aids. They also may be influenced by the longer session and more comprehensive build of the HERA C7 session. Crew Resources: The mean and median ratings for the Supply of Crew Resources are similar for both studies, possibly indicating that the design changes had no notable effect on the availability of crew resources. Crew Understanding: The mean rating for Crew Understanding is over half a point higher, and the median rating is over one point higher for the lab study than for the HERA C7 study. This difference may indicate that the modified 3D model provided participants with an advantage relative to the original 3D model with 2D diagrams. This difference also may be influenced by the need for additional knowledge in the HERA C7 study to build the rover base as well as the gripper. The Bedford workload scale was used to assess workload at the end of each assembly task. Participants in HERA C7 reported a median workload of 4, slightly higher than the median workload of 3 reported in the Lab Study. Differences in the visual aids between the two studies do not appear to notably impact subjective workload. The Usability Scale (SUS) is used to assess usability at the end of each assembly task. Participants in the VITA Lab Study rated the usability of VITA with the modified 3D model near or just above 80. Participants in the HERA C7 study rated the usability of VITA with the original 3D model with 2D diagrams just below 80. Thus, the changes in the 3D model do not appear to affect usability in any notable way.
Bibliography Type:	Description: (Last Updated: 03/25/2025)
Abstracts for Journals and Proceedings	Schreckenghost D, Holden K, Milam T, Robertson IWT, Adil K. "Enhancing situation awareness of automated procedures using adaptive multimodal augmented reality displays." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025